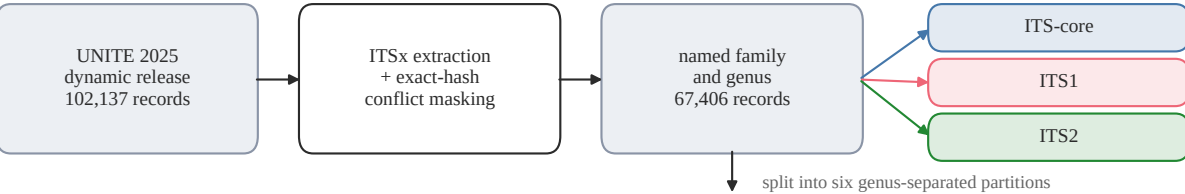
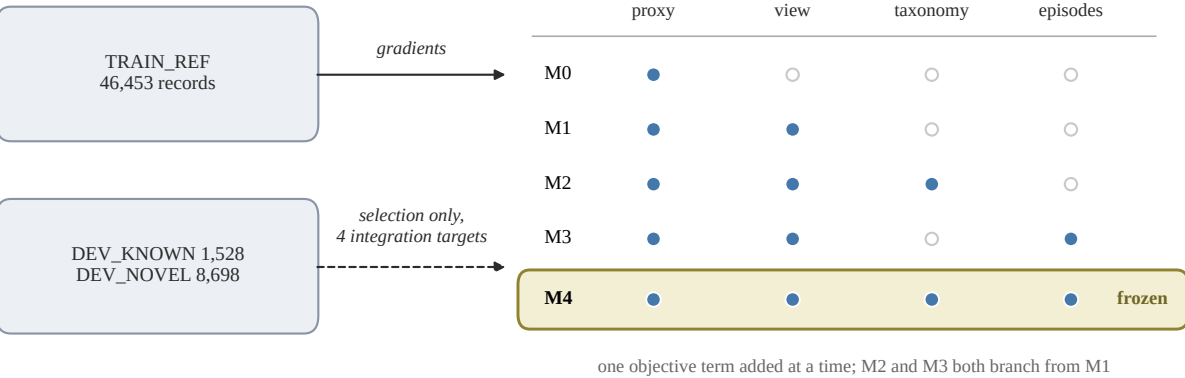


a Collection and barcode views



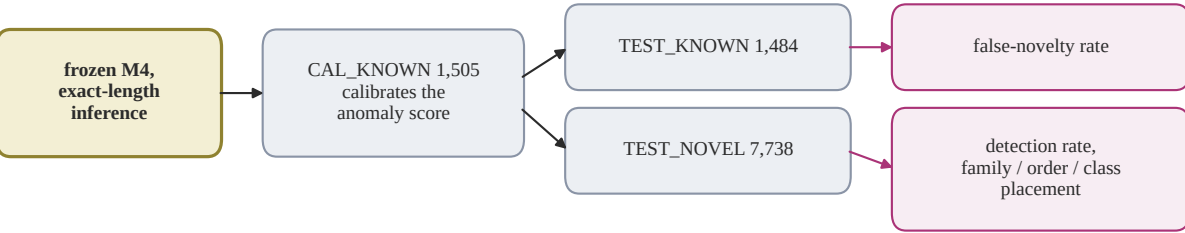
b Development: only TRAIN_REF receives gradients



EVALUATION FIREWALL · sealed by SHA-256 · CAL and TEST opened once, after freezing

c Evaluation 1 (primary): frozen M4 against a matched alignment baseline

the same queries are also scored by exhaustive VSEARCH



d Evaluation 2 (secondary): historical ITS2 benchmark

the frozen weights were never scored here; only the fixed recipe was reused

